

TO: Members of E55/E55.04

Please find enclosed a proposed standard for continuous biomanufacture. This is timely as the draft for ICH Q13 was released recently and we can take advantage of its release to align the two documents.

There are a number of reasons for having a separate document from the existing standard for continuous manufacture, E-2968. Biomanufacturing processes are complex and there are fundamental differences and challenges in their implementation, such as the need to protect the process from biocontamination, the broad scope from cell line expansion, production in a bioreactor, through purification to final fill finish.

The manufacturing approach is also dependent on modality, this standard is based on a monoclonal antibody (MAb) process. MAB's are a common therapeutic modality, they are amenable to so-called platform processes, and considerable progress is already under way in developing continuous processes. It is at the same time worth noting that many upstream processes have been run in a continuous (Perfusion) mode for some time, notably for the production of labile molecules.

Given the extended length of a continuous manufacturing process it is also prudent to consider implementing the move to continuous in a step-wise manner. This is **onsistent** with ICH Q13, and will mean that there will be situations where processes are hybrids of 'connected' and truly 'continuous' unit operations. There are also limitations on how long certain processes can be run, such as filtration, before equipment can be replaced.

This standard is intended for the manufacture of bulk drug substance. Drug product for biologics is increasingly complex; filled vials are being replaced by increasingly sophisticated alternatives such as pre-filled syringes, autoinjectors and on-body devices.

Single-use technology is enabling for continuous processes, in particular it provides a higher assurance of sterility and the avoidance of cross-contamination.

This standard does not cover all operations in a continuous manufacturing process in detail. It is recognized that there are areas such as alternating tangential flow filtration (ATF), periodic countercurrent chromatography (PCC), simulated moving beds (SMB) and viral inactivation which would benefit from dedicated standards.

(E1) -XXXX

Designation: E-XXXX-XX

Standard Guide for Application of Continuous Processing-Manufacturing (BioCM) in the Biopharmaceutical Industry¹

This standard is issued under the fixed designation EXXXX; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (°) indicates an editorial change since the last revision or reapproval.

1. Scope

1.1 This guide provides key concepts and principles to assist in the appropriate selection, development and operation of continuous processing technologies for the manufacture of biologically derived products. Although selected concepts covered here can be applied to non-biopharmaceutical products, a separate standard, (E2968 Application of Continuous Manufacturing (CM) in the Pharmaceutical Industry) is available.

1.2 Processes for biologically derived products differ from those for synthetic drugs in a number of fundamental ways in addition to their source (e.g. format – aqueous liquids vs. powders; scope - genesis to final formulation.).

1.3 Biologically derived products also differ widely from each other in terms of modalities, source materials and the processing technologies used, not all of which are equally amenable to operating in a continuous mode.

1.4 Opportunities do exist for the introduction of continuous technologies, for example, efforts are ongoing to adapt processes for large scale manufacture of broadly applicable modalities such as monoclonal antibodies to a continuous format. This standard is intended to provide guidance to the design and implementation of antibody processes. For further information, see Appendix I and references 1-3.

1.5 The principles can be applicable to unit operations and/or processes for other modalities, but may not be applicable to all bioprocesses. For further information see Appendix I and references 1-3.

1.6 Particular consideration should be given to the development and application of the appropriate scientific understanding and engineering principles that differentiate continuous manufacture from traditional batch manufacturing.

1.7 Since much of the processing is done under conditions amenable to microbial growth, maintaining process streams free from external biological impurities and microbial contamination (e.g. bioburden, viruses, mycoplasma_a) is critical.

1.8 This guide is intended to apply in both the development of a new process, or the redesign of an existing one.

1.9 A manufacturer may choose to implement continuous processing for discrete unit operations in stages as they develop process understanding prior to a fully connected or continuous process.

¹ This guide is under the jurisdiction of ASTM Committee E55 on Manufacture of Pharmaceutical and Biopharmaceutical Products and is the direct responsibility of Subcommittee E55.01 on Process Understanding and PAT System Management, Implementation and Practice.
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1.10 All values are stated in SI units. No other units of measurement are included in this standard.

1.11 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.*

2. Referenced Documents

2.1 ASTM Standards:¹

[E2363 Terminology Relating to Process Analytical Technology in the Pharmaceutical Industry](#)

[E2475 Guide for Process Understanding Related to Pharmaceutical Manufacture and Control](#)

[E2537 Guide for Application of Continuous Quality Verification to Pharmaceutical and Biopharmaceutical Manufacturing](#)

[E2898 Guide for Risk-Based Validation of Analytical Methods for PAT Applications](#)

[E2968-14 Standard Guide for the Application of Continuous Processing in the Pharmaceutical Industry](#)

[E3051 Standard Guide for Specification, Design, Verification, and Application of Single-Use Systems in Pharmaceutical and Biopharmaceutical Manufacturing](#)

[E3077-17e2 Standard Guide for Raw Material eData Transfer from Material Suppliers to Pharmaceutical & Biopharmaceutical Manufacturers](#)

[E3231-19 Standard Guide for Cell Culture Growth Assessment of Single-Use Material](#)

[E3244-20 Standard Practice for Integrity Assurance and Testing of Single-Use Systems](#)

2.2 FDA Documents:²

[FDA Guidance for Industry PAT A Framework for Innovative Pharmaceutical Development, Manufacturing, and Quality Assurance \(2004\)](#)

FDA Guidance for Industry, Process Validation: General Practices and Principles, rev1, 2011

[Quality Considerations for Continuous Manufacturing Guidance for Industry \(2019\)](#)

2.3 Other Guidance and references

- Points to Consider for Design and Control of Continuous Bioprocessing Kaltenbrunner, O in Biopharmaceutical Processing: Development, Design and Implementation of Manufacturing Processes. (2018) 741 - 754 Jagschies, G., Lindskog, E., Łacki, K and Galliher, P (eds) Elsevier, Radarweg 29, PO Box 211 1000AE Amsterdam, Netherlands.
- A Common Framework for Integrated and Continuous Biomanufacturing Coffman, J., Brower, M., Connell-Crowley, L., Deldari, S., Farid, S.S., Howrowski, B., Patil, U., Pollard, D., Qadan, M., Rose, S., Schaefer, E., Shultz, J. Biotechnol. Bioeng. (2021) doi: 10.1002/bit.27960
- The Current Scientific and Regulatory Landscape in Advancing Integrated Continuous Biopharmaceutical Manufacturing. Fisher, A.C., Kanga, M-H., Agarabi, C., Borson, K., Lee, S.J. and Yoon, S Trends in Biotechnol., (2019) 37, 253 – 267
- Continuous Downstream Processing for Biomanufacturing. An Industry Review (2019) Biophorum Operations Group Ltd.

¹ For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

² Available from Food and Drug Administration (FDA), 10903 New Hampshire Ave., Silver Spring, MD 20993-0002, <http://www.fda.gov>.

5. USP <1043> Ancillary Materials for Cell, Gene, and Tissue-Engineered Products

6. ISO 20399 – Biotechnology – Ancillary materials present during the production of cellular therapeutic products

6-7. 7. ICH Harmonised Guideline. Continuous Manufacturing of Drug Substances and Drug Products Q13 Draft Version 2021

3. Terminology

3.1 Definitions:

3.1.1 For general definitions, refer to Terminology E2363 and Guides E2537 and E2475. For definitions specific to continuous processing, refer to E2968 – 14. The following section provides clarification of how they are applied to bioprocesses.

3.2 Definitions of Terms Specific to This Standard:

3.2.1 *back mixed process*—a process with a residence time distribution (RTD) whose breadth is potentially significant compared to the mean residence time.

3.2.1.1 *Discussion*—Certain steps in bioprocessing are fully back mixed (e.g. in the case of a bioreactor or for a pooled load to a subsequent step) and quantities of material will be mixed into a single homogeneous condition such that a rapid step change in the properties of inlet material will not result in an equivalent step change in the properties of the output material, but will be reflected in a more gradual change. The rate of this change will depend on the equipment characteristics, residence volume, and the residence time distribution/degree of mixing. A fully back mixed process may be considered and modeled as one or more continuously stirred tank reactors (CSTR).

The process to produce a biologic product may contain both continuous processing steps and non-continuous steps. For example, the process to make product can be based on a continuous process, where the cells are growing continually in the bioreactor while media and nutrients are pumped into the vessel, and cells and product are removed from the bioreactor. The downstream separation of the cells from the media, and each of the subsequent purification steps may be run in batch mode or in a continuous mode. If the subsequent steps of the continuous process step are run in a batch mode, then the product from the continuous process is collected over time. This material is fed to the subsequent batch step within pre-defined ranges for collection, mixing, and hold conditions which assure the solution's stability over the time the material is collected.

3.2.2 *dynamic process control system*—an automated control system which monitors the condition of the product or the process, or both; predicts or detects a change to the process indicators or product quality away from a target condition; and then changes the process conditions during processing in order to maintain the product quality at the target value (or within the specified range of target values). An example in a continuous process producing a biologic would be controlling the cell density within a specific range in a perfusion bioreactor by continuous cell removal (“cell bleed”). Maintaining the density within that range provides for a higher assurance that the growth rate is constant, the productivity is similar, and the material produced has the desired characteristics.

Depending on the dynamics of the process step, the corrections may be applied immediately as a step change or as a time dependent function (for example, a ramp or exponential function). Such real time control systems may include feedback, feed forward control or:

3.2.2.1 *multivariate model-based control*—measurements of one or more product attributes and process conditions are used in a mathematical model of the process or process step to determine the process conditions required to achieve the desired outcome

depending on the operational objective (e.g. cell viability, product titer, purity) and process parameters are adjusted as needed based on the output from the model (i.e. dynamic control element).

3.2.3 *continuous manufacturing(CM) or manufacturing step/unit operation*—a process where, during normal operation, materials are continuously fed into the system at the same time as acceptable intermediate or product is continuously removed from the system.

3.2.3.1 Discussion:

(1) In a CM process or process step, the degree of transformation of any specific quantity of material from an initial condition into the subsequent condition is a function of the process parameters applied and either:

- (a) The position of the material as it flows through the process,
- (b) The duration that the material has been within the process, or
- (c) A combination of both (a) and (b).

(2) A CM process or process step may be operated to transform a pre-defined quantity of material into a product with pre-defined quality attributes which is then subjected to either a disposition decision or a decision of the suitability based on the characteristics of the in-process material. The size of the resulting batch can be defined in terms of one of the following:

- Quantity of output material
- Quantity of input material
- Run time at a defined mass flow rate

Other approaches to define batch size can also be considered, if scientifically justified based on the characteristics of the CM process. A batch size can also be defined as a range, e.g. by defining and minimum and maximum run time.

(3) , A CM process may be operated for an extended period of time without pre-defining the quantity of incoming materials. The quality and quantities of intermediate or finished product are defined during the operation of the process in a flexible way, based on principles of science and risk (for example, as any entity produced in a certain time, or containing a certain lot of a starting material), and subjected to a disposition decision. Note that in the case of bioprocesses, performance should be checked periodically to ensure that there are no changes over time. For example, cell line expression level or product characteristics may change as the cell line ages, or if the cell line is contaminated with another organism, such as mycoplasma or a virus. In the case of a purification step, the resin or membranes used should be cleaned or replaced periodically to avoid build up of material (fouling), which may change the properties or yield of the continuous processing step. Minor changes in process attributes during the lifespan of a continuous manufacturing operation are acceptable as long as they remain within pre-established acceptance criteria.

(4) A process consisting of a series of interconnected unit operations or transformations can be considered to be continuous even if it also contains transformations of defined quantities of material which, might be considered to be composed of a sequence of discrete events. An example of this is a continuous column purification process, where multiple columns run simultaneously, but each may be at a different stage (e.g. loading product, washing, elution, regeneration and cleaning), independently of the other columns, such that a continuous or semi-continuous flow into the column purification step and out of the column purification step may be enabled.

(5) During periods of start-up, shutdown or processing of small quantities of material, or both (e.g. for development/experimental or clinical studies), it is possible that not all unit operations within a continuous production line will

be in normal or steady state conditions at the same time, for example, start-up and shut-down of production cell culture. This condition should not automatically invalidate the definition of the process as representative of normal continuous operation.

3.2.4 process control setpoint—a process control setpoint is a specific target value for a process parameter or product attribute which is used by a dynamic control system. The dynamic process control system will determine what corrective control action to apply in order to bring the specific parameter or attribute closer to the Setpoint value.

3.2.4.1 Discussion—A Setpoint may be specified together with upper and lower target values such that corrective control action may be reduced once the value is within the specified range. A target range specified by upper and lower target values only has no explicit specified Setpoint value and hence corrective process control action is often suspended once the parameter or attribute is within the target range.

3.2.8 process disturbance—an un-requested and uncontrolled change in a measured or unmeasured parameter which has the effect of changing the process conditions or product quality (that is, a short-term transient condition).

3.2.9 process time constant—a measure of the rate at which the process can change from steady state operation at one condition to steady state operation at another condition.

3.2.10 recipe-based process control system—an automated control system which maintains specific process parameters at pre-specified fixed values (that is, according to a predetermined recipe) without adjustment of process parameters based on either measurement and feedback of product quality attributes or measurement and feed-forward of input material quality attributes or upstream conditions.

3.2.11 steady state—consistent operation over a period of time where all relevant process parameters and product qualities are not subject to variation outside of a defined range of values.

3.2.11.1 Discussion

1. Steady state implies that the process is not subject to significant variance with respect to time.
2. Achieving or maintaining acceptable product quality may require an adjustment of target values and hence a transition between two steady state conditions.
3. Process parameters can vary within a specified range in a steady state process
4. While critical parameters must remain within a specified tolerance, other non-essential parameters may exhibit process drift.

3.2.13 transient conditions—conditions where the process is disturbed from steady state or is in transition between one steady state condition to another. Transients may be due to either external disturbances, intentional changes in the selected operating conditions as material moves from one unit operation to another or to normal process variability over time (i.e., changes related to cell age, fouling, etc.). Development of a continuous process should include mitigations to prevent process-related transient conditions from unacceptably affecting product quality. Examples which can result in transient conditions are changes in raw material batches, fouling, a temperature shift in a cell culture process, or a change in the product quality attributes from one process step to another over time.

3.2.14 residence time—the time that process material is in a specific process environment/vessel/unit operation.

4. Significance and Use

This guide focuses on upstream and downstream processes for biopharmaceutical products, with a particular focus on antibody production processes.

4.1 Bioprocesses traditionally consist of discrete unit operations labeled as upstream, downstream and fill/finish operations. The objectives at each stage are significantly different, as are the operating parameters and control processes, which can make complete integration impractical (Appendix I). This guide does not imply that complete integration is a pre-requisite.

Upstream processes. The purpose of upstream processes is to generate sufficient product to meet patient requirements, preferably in the fewest number of batches. This starts with increasing biomass (cell-line expansion) to a production bioreactor, where the focus shifts to producing product. The material within a bioreactor during extended growth is heterogenous, e.g. cells will differ in age, there may be genetic drift, secreted product can differ in the residence time spent in the bioreactor, cell debris accumulates throughout the process

Downstream processes. The purpose of downstream processes is to harvest product and purify it from process and product related impurities (cell debris, nucleic acids, misfolds) to the desired level. Solids are first separated from solutes, which are then separated from each other. Certain processes may at best be semi-continuous, and some steps may be prone to fouling, which may require manual intervention.

Fill/finish operations. The purpose of fill/finish operations is to formulate the purified product in a form that ensures stability and sterility and provide a dosage form consistent with the desired product profile. They may also include inclusion in a delivery system as a combination product which is out of scope for this document.

4.2 This guide does not do the following:

4.2.1 Suggest that CM is suitable for the manufacture of all biopharmaceutical products and processes.

4.2.2 Provide guidance on issues related to the safe operation of a CM process or continuous bioprocessing equipment. It is the responsibility of the user of this standard to establish appropriate health and safety practices and determine the applicability of regulatory limitations prior to use.

4.2.3 Recommend particular designs or operating regimes for CM.

5. Operation of Continuous Manufacturing Systems

5.1 *Operational Considerations:*

5.1.1 In order to successfully introduce CM, due consideration should first be given to the overall operation and support of the system during the lifecycle of the plant and product, for example:

5.1.1.1 Considerations for process and product development:

(1) *Use qualified models of the continuous process or process step* to understand the variables which need to be controlled to produce a consistent product and to produce small quantities of material under different operating conditions during the development of the product. Understanding the effect of process variables on product properties provides a basis for designing large scale manufacturing processes.

(2) Within the qualified models, *consider the uncontrolled variables which may change over time* e.g. changes in the genetic make-up of the cells or changes in the proteins expressed by the cells over time, and how that may impact product quality. A secondary consideration is the ability to detect and characterize process perturbations such as a contamination event.

(3) Consider the suitability of the model for manufacture of variable quantities of product at stable operating conditions for clinical trials supplies.

5.1.1.2 For increasing process capacity from development to commercial production, consider the following:

- (1) Scale up of run length duration/number of cycles
- (2) Scale out by addition of parallel processing lines
- (3) Increase in production rate
- (4) A risk-based approach to increasing the scale of CM process equipment
- (5) Which parameters can be appropriately characterized using a scale down model and which parameters are not effectively replicated by the model

5.1.1.3 For stable manufacturing operations over the target run length, consider:

- (1) The ability of the system to produce consistent product over the intended duration of the operation
- (2) Key parameters, sampling points, and process limits that assure that the process is in a state of control
- (3) Mechanisms of failure and degradation of performance together with robust methods of detection. This detection will likely be dependent on the process step being monitored
- (4) Qualification of recovery procedures in the event of excursions (minor non-conformance to catastrophic failure)
- (5) Degree of redundancy in equipment and sensors required to assure continuous stable operation, and
- (6) Necessity and frequency for operator intervention in order to maintain normal operation (e.g. filter fouling).
- (7) Run to run variability in process parameters

5.1.1.4 In addition, where a site has not previously operated a continuous process, consideration should also be given to the following:

- (1) Training of development, manufacturing and quality assurance personnel in the theoretical and practical aspects of continuous processing
- (2) Impact of continuous operation on facilities, staff and systems (for example, extended shift working patterns, deviation management)
- (3) Use of single use (SU) equipment as part of the process, given that SU equipment can be supplied pre-sterilized, avoids the need for cleaning, and operates as a closed system, all of which provide a high degree of assurance of sterility (E3051).
- (4) Modification of quality systems to account for real time or near-real time rapid release of process intermediates (note that hold times may be included/optional as breakpoints, to manage the risk of product loss from an excursion further downstream).

5.2 Operating States:

5.2.1 The operation of a CM process system must be considered over the entire life cycle of the product (that is, development, validation, clinical trial supply, technology transfer, commercial manufacturing, and product discontinuation) for which it is intended to be used.

5.2.1.1 A process may begin as a series of connected steps which are integrated progressively before becoming fully continuous.

5.2.2 Risk analysis techniques, practical tests, or modeling tools, or any appropriate combination of these, should be employed to ensure that all potential impacts on product quality are understood and appropriately managed over all potential operating states. For example, the following should be considered:

5.2.2.1 Equipment start-up (for example, initialization and warm up ready for processing);

5.2.2.2 Process start-up (introduction of feed materials to start processing and reaching state of control);

5.2.2.3 Normal, steady state as defined for a particular process step, and in-specification operation (that is, verified to deliver material which is suitable to be released or move to a subsequent process step). As mentioned, during this steady state operation, specific process characteristics of the cell line or the state of the equipment should be checked such that these variables remain within acceptable ranges or limits;

5.2.2.4 Transient operation during flow rate or unit operation changes or maintenance (e.g. replacement of filters as they foul);

5.2.2.5 Replenishment of feedstock materials, particularly considering the impact of any variability in raw materials, and processing aids such as media, media feeds, buffers and filter characteristics;

5.2.2.6 Process pause or hold, or diversion to a surge vessel (for example, as a result of alarm conditions).

5.2.2.7 Process shutdown (including extracting product that meets specification);

5.2.2.8 Emptying of equipment or rejection of any residual material that does not or would not meet specification;

5.2.2.9 Cleaning/sanitization/ product/ grade changeover

5.2.2.10 Controlled safe status (software-controlled safe status (SSS), hardware-controlled safe status (HSS)); and

5.2.2.11 Mechanically shut down and out of service.

In some defined circumstances, manufacturers of drug substance may reprocess, or continue to process, material held under quarantine, provided the requirements for rework/reclaim of the production material are defined in a written procedure and rework/reclaim is approved by the quality authority

5.3 Process Robustness:

5.3.1 CM and individual continual process steps may pose challenges due to behaviors of both equipment and material which may be gradual, or stochastic (e.g., changes in gene expression or cell diameter). These changes may not be easily observed during either batch processing, scaled-down models, or short test-runs of continuous processes.

5.3.2 Suitable risk analysis, practical tests in-process testing strategies and modeling techniques should be considered in order to determine and evaluate potential challenges in maintaining stable process conditions during the operation of a continuous process or a continuous process step over the full length of the required production run, and any sampling or data review as part of this risk analysis and ongoing risk management should be done in consideration of ongoing process dynamics.

5.3.3 Consideration should be given to the following:

5.3.3.1 Management of bioburden and the avoidance of contamination events, which in worst-case scenarios can result in catastrophic failure and total plant contamination. These are not unique to continuous processing, but the combination of extended

run times, longer processing times between equipment/flow path cleaning and sanitization and increased process complexity can increase the risk of an adverse event.

Periodic cleaning and sanitization cycles should be qualified, designed into processes and validated for effectiveness where the same equipment is used multiple times (i.e. not SU).

5.3.3.2 The potential for fouling and the creation of process impurities, for example:

- (1) Fouling of equipment surfaces (for example, impact on heat transfer by product binding.)
- (2) Potential impact of binding to surfaces (e.g. resins, filters) causing product and process modification overtime;
- (3) Ducts, pipes, and tubing changes over time (for example, impact on flow patterns);
- (4) Fouling of instruments and probes (for example, impact on their accuracy, etc.);
- (5) Fouling of filters and resins (for example, impact on flow and pressure of fluids)
- (6) By-products with different or undesirable characteristics, or both
- (7) Precipitation, aggregation, and encrustation and/or blocking of constrained flow paths.
- (8) Leachables which could enter the process during normal operation
- (9) Creation of aggregates as a consequence of processing steps

Note that change-out of a filter or resin can result in fluctuations in process streams (improved flow, increases in product concentration) which should be accounted for.

5.3.3.2 Changes in raw material behavior between batches/ sources/suppliers which may not be covered within existing quality control requirements, for example:

- (1) Biochemical properties of materials, and level of impurities
- (2) Electrostatic properties particularly in dried (lyophilized) biopharmaceuticals
- (3) Safety properties.

The impact may be on product quality, process performance, or both.

5.3.3.3 Impact of environmental changes on raw material and product, for example:

- (1) Temperature,
- (2) Relative humidity (RH),
- (3) age of raw material.
- (4) Light exposure (particularly plastic, single use equipment).

5.3.3.4 Changes in plant and equipment characteristics over time and with prolonged uninterrupted use, for example:

- (1) Changes in surface finish, variability in cleaning of surfaces,
- (2) Changes in clearances due to wear.
- (3) Loss of sterility due to wear or improper cleaning methods.
- (4) Creation of leaks in vessels, tubing and connectors not intended for multiple use.

5.3.4 The maximum length of time over which the process is run may be determined by monitoring specific process and product attributes or process parameters rather than by validating a single fixed length of run time.

5.3.5 Where a single unit operation within a process line is determined to be disproportionately vulnerable to degradation in performance or lack of robustness, then strategies to maximize the potential run time in order to avoid the need to stop the overall process should be considered, for example:

5.3.5.1 Periodic/rapid change out of individual items of equipment such as filters and membranes

5.3.5.2 Redundancy (particularly for cell retention devices, and membranes used in membrane-based steps), parallelization, or duplication of critical equipment elements (for example, cell concentration devices, filters, pumps, tubing, critical in-process instruments used to measure process step control parameters).

Alternatively, running the vulnerable operation at a higher speed may allow for more downtime for cleaning etc. It will depend on the relative rates of additional steps.

5.3.5.3 Process characterization to determine the degree and frequency of vulnerability of the unit operation and requirements for preventive maintenance.

5.4 Requirement for Operator Intervention:

5.4.1 Generally, a [CM](#) should be expected to operate with the minimum practical level of operator intervention.

5.4.2 Certain types of intervention should be planned for and a course of action determined in the event of e.g. leaks, or filters or resins fouling,

5.4.3 Unplanned operator intervention should be considered as a potential source of uncontrolled variability. Continued unplanned intervention may indicate a lack of process robustness, lack of understanding of the critical process variables, or uncontrolled or unmanaged variability in process conditions or raw material properties.

5.4.4 Continuous improvement tools (for example, real time statistical process control and process automation) should be used during operation to identify the causes of any unplanned operator intervention. Appropriate actions should be taken to ensure that any impact on product quality is fully understood and that the root cause of the need for intervention is eliminated.

6. Process Design in a Continuous Manufacturing process

6.1 Principles:

6.1.1 The design of a CM process step or process requires the same good process design and engineering practices used in a traditional batch process.

6.1.2 However, the design of the CM process may require the consideration of additional factors which are not as important in a batch process, such as, fouling potential of cell retention devices, product pooling and fault recovery strategies.

6.1.3 Hence, when designing a continuous processing system, consideration should be given to the process conditions experienced by the materials as they flow through the system. For example:

6.1.3.1 The overall flow rate through the process (that is, the target plant production rate).

6.1.3.2 The balance between the process and buffer capacities of each unit operation, to ensure that the desired process conditions and overall line flow rates under the required operating regimes can be achieved. Some examples are as follows:

(1) Balancing the inflow rate of media into the bioreactor, with the outflow of product, and drawing off of cells to maintain a consistent cell density in the bioreactor growing at an appropriate doubling time;

(2) The capacity of a cell removal system should be examined to ensure that it corresponds to the volume and flow rate of the cell suspension entering into the cell removal step, and

(3) the rate, flow, and concentration of an unpurified protein solution entering into a continuous purification step should be managed throughout the individual cycles within the step. The flow rate of non-loading steps (e.g., cleaning, regeneration) may be modified to balance with the duration of loading steps. This will be dependent on load concentration, load flow rate, resin dynamic binding capacity, column volume and the number of columns.

6.1.3.3 The instantaneous/peak flow rate at locations in the system where material flow may be discrete.

6.1.3.4 The flow pattern of the materials in the system (for example, plug flow versus back mixed).

6.1.3.5 The process conditions required to achieve a specific productivity within the bioreactor or specific purity within a continuous purification system.

6.1.3.6 The process time constants, reaction rates, average, maximum and minimum residence times required to achieve a specific process objective such as specific product quality (glycosylation, recovery, purity),

6.1.3.7 The relationship between material properties, process conditions and equipment design required to achieve a reliable flow of materials.

6.1.3.8 The analysis of the mass and energy balance for the system or continuous process step using process and chemical engineering principles, for example:

(1) Capacity of physical transfer systems, and

(2) Capacity of heating systems,

(3) sufficient supply of nutrients and removal of by-products for a growing cell culture.

6.1.3.9 Implementation of appropriate monitoring tools.

6.2 Process Time Constants:

6.2.1 The time available for a given bioproduct production or purification is determined by the residence time of the material in a specific process step based on the environment in the bioreactor or purification step. The focus is the time required to produce material in the desired final conditions.

6.2.2 As the material flows through a particular CM system or step, rate limiting elements within the process must be considered to ensure that, for a given flow rate, the required process end point or product attribute can be achieved within the time available. For example, the following should be considered:

6.2.2.1 Product binding and elution in a continuous purification step

6.2.2.2 Ensuring a minimum hold time at a specified condition for a viral inactivation step.

6.2.3 The time for start up and transient conditions, for a continuous cell culture process.

6.2.4 An understanding and subsequent verification of the various time constants of the process is specifically important in determining the expected behavior of the process during start up and shutdown and hence the impact on quality decisions regarding the disposition of material manufactured during this period.

6.2.5 Consideration should be given to the use of monitoring systems which determine if the required product attributes or process indicators are achieved before product is allowed to proceed to the next unit operation.

6.3 Residence Time, Residence Time Distribution, and the Degree of Back Mixing:

6.3.1 In order to characterize a continuous process, the process residence time and residence time distribution within the CM step must be understood and quantified during both start up and normal operation as well as during process disturbance and shutdown conditions. This is particularly important when the step is dynamically changing, e.g. product production within a continuous cell culture bioreactor.

6.3.2 The mixing of product within the system is important. For each continuous step, the following should be considered:

6.3.2.1 Process modeling of each continuous step, and integration with prior and following steps (as appropriate),

6.3.2.2 Validation tests using specific markers/tracers, and

6.3.2.3 Online/inline process measurement of appropriate process indicators and product attributes.

6.3.3 Two extremes of mixing are commonly identified as “plug flow” or “fully back mixed” Bioreactor processes have a high degree of mixing. For example, product that has been produced by cells in the bioreactor, but not yet removed, may be degraded by host cell impurities that are retained in the bioreactor.

6.3.4 An estimation of the RTD within the process enables an understanding of the following:

6.3.4.1 Which output material contains which input material.

6.3.4.2 Which process conditions have had an impact on a specific quantity of output material,

6.3.4.3 How minor and transient changes in feed or process conditions will impact output product attributes, and

6.3.4.4 The degree of recycle.

6.3.5 In particular, an understanding and quantification of the residence time distribution may be used to determine which material may have been affected by a deviation in process conditions and hence the specific identity of any product within the scope of any investigation or disposition decision. This understanding is particularly important in live cultures of cells, which deviations could affect the viability and or properties of the live cells, and their product. Similarly, purification continuous steps could have deviations in some variables (pH, temperature) which could cause major changes in the properties of the product.

6.4 *Product Transport and Material Properties:*

6.4.1 A CM process frequently consists of a number of connected unit operations.

6.4.2 Careful consideration should be given to the design of transport and flow control elements within a continuous system in order to ensure that materials will flow in a predictable way without adverse impact on product quality.

6.4.3 Transporting and controlling the flow rate of solutions or suspension of cells may be a specific problem in this respect. For example, rapid transport of cells to and from the cell retention device may cause cell lysis; excessive agitation may cause aggregation of product due to excessive air/liquid interface contact. Hence, the handling and flow properties of materials to be processed should be determined as early as possible within the development of the product such that the process equipment may be designed appropriately.

6.4.4 Characterization of materials using laboratory techniques on small samples may give good early indication of potential problems. But where there are concerns about material properties, it is recommended that testing of representative equipment and representative materials is carried out under representative conditions as early as possible.

6.4.5 Transport processes may also cause some degree of transformation (for example, segregation of cells or aggregation of proteins) and must be designed carefully to ensure the following:

6.4.5.1 Effects are identified and understood,

6.4.5.2 Steps are taken to minimize such effects during equipment and plant design, and

6.4.5.3 Controls are put in place to manage or mitigate such effects over the entire range of operating conditions.

6.4.6 Characterization of leachables from process equipment, especially disposable plastics and membranes, should be performed and assessed for their ability to impact product quality or process performance.

7. Product Quality Control for Continuous Manufacturing

7.1 CM provides opportunities to monitor and improve control of the critical parameters of the process and the critical quality attributes of materials in real time as materials flow through the system and as instrumentation is available.

7.2 The application of Guides E2537, E2898 and E2968 should be considered.

7.3 Risk assessment and process understanding should be employed to determine the degree and type of monitoring and process control required to produce material within specification and should take account of the feasibility of implementing reliable control action.

7.4 A real time flow of process and product information from a well-designed CM process or process step provides an opportunity to gather more information which will aid in process characterization and further advancing process understanding.

7.5 Sampling and Data Collection within a CM Process:

7.5.1 The timing and appropriateness of measurements used to monitor and control a continuous process or process step, will depend on the purpose and the nature of the step (e.g. product titer in a bioreactor vs impurity profile further downstream) as well as the following characteristics

7.5.1.1 The intended scale of scrutiny of the sampling and measurement system;

7.5.1.2 The impact of product flow on product characteristics;

7.5.1.3 The extent that measurement made is representative of the complete process stream;

7.5.1.4 The impact of process dynamics on the requirement of the sampling and measurement system, which may be limiting on the flow rate at which a continuous unit operation can be run. The impact of parameters which may vary within that process step on quality attributes and/or process performance should be understood, so that the relationship between changes, measurements and effective response times are harmonized with each other

7.5.2 The frequency of measurements should be appropriate to the dynamic response time of the parameter or attribute.

7.5.3 Due consideration should be given to how to handle measurement noise due to variability in sample presentation which is potentially greater in an online system compared to an at-line or offline system, where the sample presentation may be better controlled.

7.5.4 Specific consideration should be given to the following:

7.5.4.1 The tradeoff between measurement quality and measurement frequency,

7.5.4.2 The use of filtering to remove signal noise,

7.5.4.3 The impact of filtering on dynamic response of the measurement (that is, loss of dynamic response),

7.5.4.4 The use of suitable signal processing or statistical techniques in order to extract meaningful process information from background noise, and

7.5.4.5 Strategies for avoiding excessive control action in response to normal process variability.

Robustness of Instruments and Analyzers

7.6 Information Used for Product or Process Control May be Derived From a number of areas including the following:

7.6.1 Direct measurements of Critical Process Parameters (CPPs) such as temperature, pressure, cell density, pH, ionic strength, osmolality and flow rates that may directly change the quality of the product if out of the defined range.

7.6.2 Direct measurements of Critical Quality Attributes (CQA)s via suitable online or at-line analytical technology, for example,

7.6.2.1 Spectroscopic or mass spectroscopy analysis for composition

7.6.2.2 On-line or at-line HPLC/UPLC analysis of product variants and aggregates.

7.6.3 Predicted values for a CQA using a combination of one or more CPP(s) in a validated model. This may be particularly useful where a CQA may not be measured directly during the process, such as viral inactivation at low pH.

7.7 In all cases, it is essential that the primary sensors are robust, reliable, and accurate over the expected run time of the process and likely range of the parameter to be measured. In-line sensors should be calibrated and measurements periodically verified off-line. Consideration should be given to:

The long term effects of fouling on a sensor

The effects of short and long term sensor drift (e.g. lamp intensity)

Requirements for cleaning, recalibration or maintenance

The impact on the process of complete failure

The possible need for duplicate or redundant sensors

The strategy for reconciling potentially different values from multiple sensors or alternative sources of data

The strategy for ongoing validation of sensors

7.8 Use of Surrogate Measurements:

7.8.1 When the direct measurement of specific process parameters or material attributes cannot be made, then surrogate measurements of the product or process which can be taken and used in conjunction with suitable models may be considered. Consideration should be given to the degree of correlation between the surrogate and direct measurement

Material Sourcing and Variability

7.9 Principles

7.9.1 Materials used in bioprocesses are exceedingly diverse. It is essential to use well characterized materials from qualified sources in validated processes.

7.9.2 Materials range in complexity from simple (frequently compendial) materials such as salts and buffers, more complex nutrients and growth factors (some of which may also have biopharmaceutical properties), processing aids such as resins and filters, and contact materials such as tubing manifolds, valves and single-use systems.

7.9.3 Material sources are diverse. Materials may be mined, synthesized, or extracted from natural sources. It is critical to understand the source in order to determine risk (e.g. viral safety for animal sourced materials).

7.9.4 Many materials will have undergone extensive processing during their manufacture (e.g. heat, gamma irradiation) and it is important to understand how the method(s) of manufacture can contribute to variability which may impact product quality or process performance.

7.9.5 Many materials are not intended to be incorporated into the final product. Such materials may be referred to as ancillary materials: (USP <1043>, ISO 20399). Cell culture materials may function as nutrients and be incorporated into product but any excess nutrients not consumed are diluted and removed during extensive purification.

7.9.6. Materials should be assessed in a science- and risk-based approach. Risk assessments should be documented and communicated and risks should be managed through an approved plan, which may involve e.g. qualification of alternate sources, suppliers or sub-suppliers, inventory management, or process modifications, depending on the nature of the material(s) and the risk(s) posed.

7.9.7. Suppliers should be assessed based on their quality systems, technical capabilities and commercial stability (e.g. ASTM Standard E3051 provides guidance for suppliers of single-use materials).

7.9.8 Materials should be fit for their intended purpose, with the oversight of a robust quality management system. It is recognized that this will also be a function of the nature of the material and its source; for example ancillary materials may not be manufactured to cGMP standards, in which case other quality systems may be acceptable (ISO 20399)..

Material Identification and Traceability

In a CM process, the producer must ensure material traceability for each batch manufactured, which is critical in enabling a product recall should that prove necessary.

It is also necessary to track material between the location at which the control decision is made and the point at which material is removed.

If a process is operated within a state of control on an ongoing basis, it could be possible to designate large quantities of product to be of uniform quality, even though different batches of raw material and/or different processing conditions may have been utilized during the production run.

7.10 Principles:

7.10.1 Before implementing a CM process, a discontinuous stream of specific lots of raw materials must be evaluated and released for use. It must be possible to identify lots of incoming raw materials and reliably link the relevant process information to these lots in a timely manner.

7.10.2 This requires a demonstrated ability to do the following:

7.10.2.1 Identify the identity, quality, and quantity of raw material, intermediate or final product, or both.

7.10.2.2 Understand which materials are intended to be incorporated into the final product and which materials are present as ancillary materials

7.10.2.3 Record when and where specific quantities of particular lots of raw materials enter the process. N.B. chemically defined media used in cell culture are highly complex and keeping track of which lots of component chemicals are present in each lot of complete media requires a high degree of transparency and communication between suppliers and manufacturers. Electronic exchange of information is highly advantageous (ASTM standard E-3060-17)

7.10.2.4 Record where and when specific quantities of particular lots of materials exit from the process.

7.10.2.5 Characterize the CM process or process step to be able to determine how the flow patterns (for example, residence time and residence time distribution) at different flow rates and operating conditions may impact material flow within the system, especially within the limits material is collected for further processing or release. For example, in a continuous cell culture system, the growth rate of the cells at various cell densities could dramatically change the production rate of the product, the residence time in the bioreactor, and may change the characteristics of the product. This may also be influenced by the replacement rate of the media. The effect of variables that can effect product quality should be identified before a continuous process step is finalized.

7.10.2.6 Characterize the CM process or process step to be able to determine how the dynamic behavior of the process at different and varying flow rates or operating conditions, or both, may impact process performance, especially within the limits that process intermediates or product are collected for release. In the case of a continuous bind and elute purification step, the contact time of the product stream with the resin is critical in determining the amount of product bound to the resin; similarly, the tightness of the binding to the resin and the time required for the product to escape from the resin particle upon elution will affect the purity and recovery of product.

7.10.2.7. Bind and elute chromatography steps require multi-column set-ups in order to run in a continuous mode, cycling through equilibration, sample loading, washing, elution regeneration and periodic cleaning and sanitization steps. It is important that the pressure/flow rate characteristics of such columns are highly similar to ensure consistent flow properties and performance across multiple steps.

7.10.2.8 Understand the overall flow of intermediates and product in the system or subsections of the system (that is, mass balance) and the ability to account for material which may be affected by the following:

- (1) Removed deliberately from the system for sampling;
- (2) Unintentionally lost from the system due unforeseen events;
- (3) Retained in the system, for example, in the form of stable surface coatings or fouling;
- (4) Held up in the system and subsequently released on a controlled basis, for example, material held on filters and subsequently released on filter blowback;
- (5) Held up in the system and subsequently released on an uncontrolled basis, for example, hang ups in vessels, etc.; and
- (6) Removed from the system but subsequently determined as being of suitable quality and re introduced to the system for processing.

7.10.2.9 Demonstrate a reliable and timely link between relevant product quality information (for example, values of CPPs and CQAs) and any specifically identified product, taking into account all of the above considerations.

7.10.2.10 Understand and account for the effect of differing lots of raw material on equipment calibration.

QUALITY CONSIDERATIONS

7.11 Principles

7.11.1 Regulatory agencies have endorsed continuous manufacturing principles for biologic products (2). Evolving from a batch to continuous manufacturing process for biologics may require new and innovative quality practices to match the new manufacturing processes. Quality practices that approve and release raw materials, process intermediates, and final product may need to be adjusted from previously established batch processing to fit the technology. The use of Process Analytical Technology (PAT) greatly facilitates the implementation of continuous processing.

7.11.2 A continuous process step may take incoming material streams from discontinuous sources (i.e. different lots of chemicals) or from another continuous workflow. The output can be discrete lots of material or used as the feed to another continuous unit operation. How in process lots are defined will depend upon the individual unit operation, and on the batch-definition strategy of the manufacturer

7.11.3 A quantity of material (i.e. a “lot”) subject to a quality disposition decision could be defined as:

7.11.3.1 All of the material discharged from the process between two specific times (irrespective of the amount of material produced).

7.11.3.2 A specific quantity of material produced (irrespective of the time taken).

7.11.3.3 All of the material produced between two specific process events (for example, specific process condition changes).

7.11.3.4 All of the material that is “intended” to contain a specific lot or quantity of a specified input material.

7.12 Continuous processing systems may be designed such that a portion of in-process material or final product that does not meet quality requirements is diverted for re-processing or to waste

7.13 As a consequence, it is possible that not all of the materials that were originally fed into the continuous process step or continuous process, as part of the original single manufacturing order, will be in the finished product intended for release to the market.

7.14 Regulations may define the quantity of product for release to the market (that is, ‘batch’) in a way which may impact on the approach taken to segregate good product from out of specification material. For example, a batch can be characterized as a defined quantity of product that meets the product specifications and attributes, a manufacturing process with a continuous process or process steps may produce multiple batches.

7.15 Hence where the diversion/separation of material which does not meet quality requirements is employed as part of the product quality control strategy then this must be carefully assessed and may need to be justified to the appropriate regulatory authority.

7.16 Diversion/acceptance criteria must be suitable to ensure compliance of the entirety of the material subjected to the release decision (“good material”) to the applicable specifications.

7.17 Real time diversion/rejection of material which does not meet in-process specifications must be justified by proper demonstration that the diversion/rejection decisions are based on reliable data and proper understanding of process dynamics. The goal here is to ensure that all of the in-process material that is manufactured continuously meets all quality attributes of safety, efficacy, purity, potency, etc. thereby ensuring patient safety. Courses of action may include:

7.17.1 Terminating the campaign

7.17.2 Rejecting affected portions of the material in the process until an affected parameter comes back into tolerance. In this situation, there will be a transient condition whereby the in-process parameter transitions from an out-of-specification condition to a within-specification condition. The timing as to when the in-process material is deemed acceptable must be justified. Re-processing the rejected material may be acceptable with appropriate justification.

7.17.3 Investigating the reason(s) why the parameter is not within specification to determine the root cause of the event.

7.17.4 Investigating the effects of the out-of-specification condition on process equipment, purifications resins, etc.

7.18 Excessive rejection of material may indicate that the underlying process is not robust or is not operating normally, or both.

7.19 Appropriate process engineering, modeling, testing and risk assessment may be required to ensure that the principles behind the following elements are well understood.

7.19.1 unique identities will be assigned to quantities of final product leaving the process, including the following:

7.19.2 specific quantity of final product will be given a unique identity.

7.19.2.1 consideration is given to risks associated with producing large quantities of product over a long production run, (particularly considering the risk of contamination from difficult to detect virus contaminations or other biological contaminations, or changes in the cell productivity due to gene changes in the majority of cells) and

7.19.2.2 Product that may contain multiple lots of input materials.

8. Process Control Systems for Continuous Manufacturing

DESIGN

8.1 *Design Objective:*

8.1.1 The design of the process control for a continuous process or continuous process step should consider the following objectives:

8.1.1.1 Ensuring, with a high degree of probability, that all material released from the process will meet the required specification limits.

8.1.1.2 Reducing variability within the acceptable specification limits by automatically taking appropriate actions in a timely manner to adjust process parameters within a validated range such that the product quality is maintained closer to a specified target or set point.

8.1.1.3 Maximizing productivity and minimizing waste.

8.2 *Design Principles:*

8.2.1 Objective (8.1.1.1) may be realized by an automated recipe-based control system with appropriate sampling, testing, quality control procedures, and equipment mechanisms to detect and reject materials which are out of specification. Objectives (8.1.1.2) and (8.1.1.3) are generally more likely to be achieved by a dynamic process control system

8.2.2 The effectiveness of a particular control strategy for a continuous process step and the requirement for dynamic control action can be determined by using suitable statistical tools. For example, Statistical Process Control (SPC) that may be used in real time to do the following:

8.2.2.1 Determine the robustness of the process and normal variability of product quality achievable by the process.

8.2.2.2 Quantify the normal capability of the process to consistently produce good quality product.

8.2.2.3 Identify disturbances which might lead to the production of out of specification material if no positive control action is taken.

8.2.2.4 Identify the improvements in process robustness and capability due to the application of dynamic control action.

8.3 *Design Considerations:*

8.3.1 The design of a control system for a continuous process step should be considered in terms of the following:

8.3.1.1 Ease of initial development and validation.

8.3.1.2 Potential to produce material of consistent product quality.

8.3.1.3 Availability of suitably accurate and reliable sensors.

8.3.1.4 Ability to develop suitable control strategies which can respond predictably to sources of variability (including variability in attributes of materials which affect the process but are currently not routinely measured).

8.3.1.5 Risk associated with permanent or transient sensor failure.

8.3.1.6 Requirements for redundancy or internal continuous verification, or both.

8.3.1.7 Process dynamics (robust process operating ranges) and product CQAs.

8.3.1.8 The ability to display in real-time and record quality data for the product and the process (ideally in the form of control charts or an equivalent).

8.3.2 For dynamic control systems, the following configurations should be considered:

8.3.2.1 Feedback control.

8.3.2.2 Feed forward control.

8.3.2.3 Multivariate model-based control.

8.3.3 The rate at which corrective actions are applied to the process must take account of the dynamic behaviors of the process. Failure to do this may result in a situation of “over control” which may result in an increase in variability as a result of excessive control action.

~~CONTROL SYSTEM ROBUSTNESS~~ Control System Robustness

8.4 *Operating Considerations:*

8.4.1 Consideration should be given to how the control system may operate during extended production runs where individual elements (for example, online analyzers) may require routine attention (for example, for cleaning, preventive maintenance, recalibration). The use of duplicate sensors and/or analyzers can allow processing to continue uninterrupted.

8.5 *Design for Robustness:*

8.5.1 Good process engineering practices (for example, quality by design, time or frequency domain analysis) should be employed to ensure robust performance

9. Development of Continuous Manufacturing Processes

9.1 The development of a continuous process or process step should follow established principles and good practices which are applied to all pharmaceutical process development projects. Additional consideration should be given to the following:

~~PARAMETER SCREENING~~ Parameter Screening

9.2 *Principles*

9.2.1 Connecting multiple unit operations together to get a [continuous manufacturing](#) stream is challenging in bioprocessing. The criteria used to monitor and control upstream and downstream processes differ significantly (e.g. growth and cell viability vs protein concentration in a purification stream). Processes may also be inherently variable or sensitive to small changes in the incoming feed streams (e.g. presence of trace metals in chemically defined media).

9.2.2 In order to address variability, each unit operation can be characterized independently outside of the continuous system using a qualified model with a representative feed stream lot.

9.2.3 Variability in the incoming feed stream should be characterized where appropriate as a process parameter.

(1) *For example* – Variability in the pH of a harvest stream in a perfusion bioreactor can affect the loading on a Protein A column if the two unit operations are connected. The Protein A step can be characterized in a representative scale-down model independently of the bioreactor using feed streams of different pH values in order to evaluate the design space.

The effects of changes to process parameters and raw material attributes on product attributes may be investigated utilizing a process characterization approach based on statistical experimental design

9.3 Both theoretical analysis and practical testing with representative materials may be used to identify the process parameters and determine ranges of values to be investigated in further studies.

9.4 Investigations should be conducted using a structured approach which is statistically valid. It is important that these investigations capture the effects of repeat execution.

9.5 Where individual process steps or unit operations are developed in isolation, the potential interaction among the individual steps should be considered when they are integrated into a fully continuous process. In particular, a unit operation/ process step that creates acceptable material in isolation could be unsuitable for extended operation in a fully integrated process line.

Determination of Process Dynamics

9.6 The process dynamics, that is, rate and sensitivity, should be understood at the level of individual unit operations, as one step may prove to be rate limiting for others in a fully connected process.

9.7 The impact of changes in the process production rate or equipment scale changes, or both, on the process dynamics should also be considered.

Process Validation and Continuous Process Verification

9.8 FDA Guidance for Industry for PAT, Process Validation and ASTM Guide [E2537](#) should be consulted for guidelines for process verification. Guide [E2898](#) should be consulted for the validation of PAT applications.

9.9 Consideration should be given to the following in verifying the ability of a continuous processing system and process control strategy to achieve a state of control and deliver the specified performance and reproducibility:

9.9.1 The process conditions which define the end of start-up and the shift to production conditions under which material can be collected and subsequently released (i.e. where product quality is within acceptable limits).

9.9.2 The ability of the process to reach and detect the start of normal production in a state of control.

9.9.3 The ability of the system to maintain the intended process conditions over the intended life of the process.

9.9.4 The ability of the system to detect excursions from the target CPP or CQA values and use a sufficient understanding of process dynamics in order to do the following:

9.9.4.1 Divert to waste identified non-conforming material

9.9.4.2 Divert to quarantine for subsequent analysis any identified potential non-conforming material, and

9.9.4.3 Shutdown the process.

9.4.5 After diverting material to drain for a temporary period of time, adequate justification needs to be provided in order to initiate the collection of material when product quality is acceptable.

Cleaning Process and Cleaning Verification

9.10 The cleaning process and frequency of cleaning of multi-use equipment should be qualified and the effectiveness verified periodically.

9.11 The design and verification of the cleaning process should consider:

9.11.1 Design for worst-case soils

9.11.2 Material holdup

9.11.3 Degradation of the material within the process

9.11.4 Microbiological growth

9.11.5 Formation of chemical and biologic films

9.11.6 The advantages and practicality of clean in place versus off-line cleaning

9.11.7 Ability to remove cleaning agents

9.11.8 Required level of cleanliness, and ability to monitor state of cleanliness and/or to detect contamination

9.11.9 Replacement frequency of single use and sterilized components

9.12 The cleaning frequency may be defined in terms of:

9.12.1 Elapsed operating time

9.12.2 Quantity of material processed

9.12.3 History of process conditions or deviations.

9.12.4 Sterilization time and expiration of single-use or sterilized components.

10. Specifications of Continuous Manufacturing Systems

Plant Requirements

10.1 The supporting functions of a facility implementing a continuous CM process system should follow good pharmaceutical engineering practices.

10.2 Solution and water management is a significant issue for upstream media and downstream buffer preparation (1) and consideration must be given to the space and resources required for solvation/hydration of solid raw materials and/or the use of concentrates.

10.3 For a continuous process step, particular consideration should be given to ensuring that the user requirement covers the following:

10.3.1 The required operating pattern of the plant, for example:

10.3.1.1 Operating hours and FTEs to operate during the operating hours,

10.3.1.2 Change over requirements

10.3.4 The expected nature of the input, intermediate, and output materials.

10.3.4.3 Exposure or other hazards, or any combination thereof (for example, explosion and contamination risks), and

10.3.4.4 Sensitivities to degradation of product based on known variables which vary (pH, osmolality, air/liquid interface creation, etc.).

10.3.5 The transformation steps required and associated kinetics,;

10.3.6 Expected requirements for material traceability, for example:

- Expected methods for product quality measurement and verification, include the following:
- Expected methods for product and process control strategies.

10.3.7 The identification and range of external factors or disturbances that can be reasonably expected to influence the process, for example environmental conditions (temperature or humidity, or both), etc.

10.3.8 Expected yield.

10.3.9 Requirements for space (notably for raw material and SU system storage, which can be considerable).

System Specification

10.4 In the specification for a CM process system for pharmaceutical manufacture, consideration should be given to allow for the following:

10.4.1 The process may move to CM in a step-wise manner (i.e. not all parts of the process are continuous from the start)

10.4.2 The process may not be fully connected from the start, allowing valuable experience of individual unit operations to be gained, OR breakpoints are designed into the process, to decrease the exposure to risk in the event of an excursion.

10.4.3 The process should be designed with the potential for changes as technology evolves (e.g. process operations should be defined in terms of functional performance to allow for changes within design space as appropriate).

10.4.4 **Single-use technology** provides certain advantages for continuous bioprocessing such as a higher assurance of sterile conditions during processing, and the potential to develop closed systems. The available single-use systems are also suited to the scale of operation. However there are also unique considerations; equipment is not designed with re-use in mind which results in manual intervention and a consequent increased risk for contamination events. Considerable demands are placed on the supplier-end-user relationship for quality systems and scheduling and there will be a significant impact on inventory requirements. ASTM Standard E 3051-2017 provides guidance for the qualification of suppliers and verification of processing equipment. Consideration should also be given to certain specific issues pertaining to the use of single-use equipment, such as compatibility with bioprocesses and leaks based on material integrity (E3231-19, E3244-20)

Fault resolution

10.5 Faults or excursions from expected performance should be analysed and understood before processing continues. This is more challenging in continuous mode than in batch processes. Material may need to be segregated and the issue documented.

10.5.1 Consideration should be given to probable issues and corrective responses should be developed in advance.

10.5.2 Responses should be commensurate with the risk(s) posed. Risk will be a function of the nature of the issue, the point in the process where it occurs and the degree of intervention required for correction

10.5.2.1 Leaks in e.g. manifolds for buffers may pose a minor risk, especially if they are upstream of bioburden filters.

10.5.2.2 Leaks in cell culture media handling systems pose a greater risk for contamination

10.5.2.3 Gross leakage/catastrophic failure can result in significant loss of product

10.5.2.4 Viral contamination can have an impact at the facility level

10.5.3 Corrective actions can include the availability of redundant equipment, the planning of identified hold points, availability of surge tanks, and qualification of re-processing procedures.

10.5.4. Consideration should be given to having a wide design space for a given process step and a wide range of responses to issues.

11. Keywords

11.1 application; continuous processing; pharmaceutical industry

APPENDIXES

(Nonmandatory Information)

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APPENDIX I

Bioprocessing is used in multiple industries in the production of pharmaceuticals, foods, flavours, fuels and chemicals with the aid of enzymes, microorganisms, plant and animal cells in a bioreactor or extracted from tissues. Even within pharmaceutical processing the range of materials and techniques used is extremely diverse, so it is not possible to describe a single best practice. Materials may be processed from animal, plant or human sources, or produced in a bioreactor from prokaryotic or eukaryotic cells. Note that perfusion technology is already established in commercial biomanufacturing.

Monoclonal antibodies are a widely used modality, and may be required in significant quantities (up to metric tonnes) depending on their mode of action. Because of their similarities they are amenable to platform approaches. Platform manufacturing approaches may require a certain amount of optimization from one molecule to the next, but they are sufficiently similar and the quantities required are sufficiently large to justify the investment in developing a continuous manufacturing approach.

This Appendix provides a brief discussion of some of the techniques that can be used in the manufacture of monoclonal antibodies (MAbs), and considerations for moving towards a continuous process. More comprehensive reviews are available in recent literature (1-3). Some additional comments are provided for other modalities where a partially or fully continuous approach may be considered. It is not intended as a comprehensive guide to bioprocessing, which would be well outside the scope of this standard.

General

To the extent to which it is possible, processes are designed in such a way that the output from one step is suitable for direct application to the next with a minimum of further conditioning.

Certain unit operations have significant mixing, so concepts of ‘plug flow’ and the movement of product through the process over time should be used as appropriate. Examples are bioreactor conditions and cross-flow or tangential filtration. Other operations, such as viral inactivation, require conditions to be held for a defined period of time, which can be accomplished in more than one way.

Since bioprocesses are run in conditions amenable to microbial growth, bioburden management is a significant overall consideration. This is particularly true for continuous operations where run times are extended significantly. Consideration should be given at the process, personnel and facility levels

Given the extended scope of a fully continuous operation, consideration should be given to designing redundancy and break points into the process, to manage risk and to allow for equipment replacement in the event of fouling (e.g. filters, chromatography columns, sensors) or leaks (which do not

necessarily equate to microbial ingress). The inclusion of break points will impact which other steps may (or may not) become rate limiting.

Upstream: Source material

Therapeutic proteins, including MABs, are typically produced in mammalian cell culture, using cells derived from Chinese hamster ovaries (CHO cells), although a wide range of other types can also be used. Cells are grown in bioreactors up to 10,000 – 25,000 liters in scale in batch operation. Moving to CM operation, with the use of perfusion technology and Alternating Tangential Flow filtration (ATF) can allow for a tenfold reduction in the size of bioreactors. Note that requirements for cell culture media remain at a minimum the same, or may even be increased. Expression levels can be in grams per liter. Other expression systems are also used, notably microbial fermentation, as well as other eukaryotic cells. The choice of source material has a profound impact on growth rates, expression levels, scale of operation, and subsequent processing steps.

Upstream: Production bioreactor

A production run starts from a cell bank; cells are inoculated into growth media and are further sub-cultured to create sufficient biomass for production scale operations. This process can take several weeks, depending on the cell line. Cells are typically grown in suspension culture, although some cells are anchorage dependent and require a solid substrate for growth.

The production bioreactor can be run in batch (fed-batch) mode, or in perfusion mode where spent media is withdrawn and cells are grown continuously for an extended period of time (up to months). The footprint required for a perfusion bioreactor is significantly less than that required for a batch process, however the volumes of media required are significant in both cases.

For perfusion operation of suspension cell culture, cells are retained by filters, or can be separated from culture media by centrifuges or gravity settling and returned to the bioreactor. Anchorage dependent cells can be bound to an immobilized matrix or which retains the cells in place while media flows around them.

A perfusion bioreactor can be used to inoculate a production reactor in batch mode, or used for production itself. In the former, cells are transferred to the production reactor and the objective is to reach a certain seeding density, in order to have predictable growth and harvest performance and to reduce low productivity time in the production bioreactor. In the continuous mode, product is removed from the bioreactor along with spent media by filtration, whilst cells are retained in the bioreactor. Note that filtration conditions can be modified to allow for retention of product until a certain product concentration is reached

Running a bioreactor in continuous mode is complex and the process requires significant optimization. The process should be modeled extensively at lab scale prior to scale up. Consideration should be given to the following :

1. Selection of cell line, physical and genetic stability
2. Media composition and design, feed strategies, material genealogy
3. Use of dry powders vs concentrates, hydration strategy, in-line dilution
4. Bioreactor design, run rates, robustness, mixing technology, aeration
5. SUS vs multi-use designs, sensors and controls
6. Microbial controls, bioburden, endotoxin, virus management
7. Filtration technology, filter selection, equipment redundancy, tangential and alternating tangential flow filtration (TFF and ATF)
8. Feed and harvest strategy

The duration of a bioreactor run over an extended period should be determined during development based on the performance of the process, impact to product quality and the ability of downstream processes to deal with product and process related impurities. Product quality can drift over time, notably if the product has post-translational modifications, and since the bioreactor is continually mixed, product molecules will spend differing amounts of time within the bioreactor and material from lysed cells (nucleic acids, lipids, enzymes) will accumulate and can result in detrimental effects to product over time.

Microbial cells are frequently used for recombinant proteins. Metabolic rates are much higher so there can be issues with temperature control (cooling) and product may or may not be secreted, which can have a radical impact on the harvest strategy..

Upstream: Harvest

Prior to further processing product has to be recovered from the bioreactor and separated from solids (cells, cell debris). This can be achieved by centrifugation and/or filtration in a fed-batch process, more typically by cell-retentive filters in a perfusion process; the challenge at this stage is the fouling of filters caused by the accumulation of debris on the filter surface. There are various approaches to managing fouling which can include using multiple filters of different grades, the use of filtration aids, centrifugation/filtration hybrids, tangential flow filtration (TFF) and alternating tangential flow (ATF). Ultimately all filters will foul and there must be a strategy for replacing them without interrupting the flow of material through the process. Alternate approaches which avoid filtration include gravity separation, acoustic wave technology and centrifugation (1)

Microbial processes may result in products which are not secreted but have to be extracted from intact cells, and solubilized. This presents additional challenges to continuous processing.

Downstream: Capture

Further processing of the clarified solution containing product is required to remove process- and product-related impurities and potential contaminants. These technologies are discussed in greater detail elsewhere (1, 2, 4).

The material from the bioreactor post clarification is in a dilute state, along with other soluble materials from the bioreactor. Product is typically captured from solution using a highly selective adsorbent (bind and elute chromatography using Protein A in the case of Mabs). This process requires that conditions are modified in discrete stages to allow for binding of product, washing to remove spent media and unbound species, elution to remove product, further washing, cleaning and re-equilibration to starting conditions. The scale of the operation has to be sufficient to meet productivity requirements (g/L/h). In order to do this in a continuous manner, multi-column arrangements are cycled through the different stages in specialized systems designed for purpose, e.g. periodic countercurrent chromatography (PCC), or simulated moving bed chromatography (SMB). The technology can be based on packed beds, fluidized slurries or specialized high molecular weight carriers and filters impermeable to the conjugate. In each case it is important that the flow rate through each of the linked steps is carefully managed in order that all steps run in a harmonized manner, any step that is rate limiting becomes rate limiting for the whole process. For packed beds this requires a high degree of reproducibility between different columns within the system. Considerations should be given to performance parameters including (but not limited to):

1. Physical parameters of resins (particle size, pore size, rigidity)
2. Ligand selectivity, dynamic binding capacity, productivity
3. Resistance to rigorous cleaning/sanitization methods
4. Reproducibility of packing, flow performance
5. System complexity, automation, use of PAT

This step also represents a meaningful point for a planned break in the process. Eluted product is in a concentrated form and free from process-related contaminants which may have a deleterious effect on product.

Viral safety

Products derived from mammalian cells are required to undergo procedures demonstrated and validated to inactivate and remove viruses. A low pH hold under defined conditions has been demonstrated in many applications to be effective in inactivating retrovirus (E2888) and is frequently positioned post-capture, since elution from the capture column is often effected by lowering pH. The defined conditions (pH, time E2888) can be accomplished either by using an extended loop (designed to ensure mixing) or alternating hold vessels. In the former case the operation will differ from a batch process and should undergo the appropriate validation for mixing and inactivation.

Inactivation can also be achieved through the use of detergent (E3042), integrating this technique into a continuous process is more challenging. Inactivation through the use of UV is used in plasma fractionation, but has not seen significant uptake in antibody fractionation.

Regulators require the use of orthogonal techniques in processes to ensure a high reduction level of virus. Nanofiltration is a widely accepted step for removal by size exclusion which can be introduced at any point later in the process once volumes are manageable and the level of impurities are lower (to improve resistance to fouling), frequently after further polishing steps (see following). Other steps are also capable of removal by adsorption to an immobilized matrix, frequently based on anion exchange.

Polishing

Subsequent purification steps remove remaining process- and product-related impurities. To the extent possible, steps are run in flow-through mode, (i.e. product passes through the adsorbent unbound, impurities are retained). Operations are not capacity limited (for product), but there will be some dispersive effects depending on the format (e.g. packed bed, membrane, fiber).

Formulation, fill/finish

The purpose of formulation is to mix the active drug substance with various other chemicals as stabilizers, preservatives, and buffers to ensure the appropriate stability, concentration and compatibility with physiological conditions prior to filling into the final drug product. Steps can include ultrafiltration/diafiltration for concentration/equilibration with formulation solution and sterilizing filtration. The same considerations should be given to filtration operations as in previous steps.

Drug substance may be stored as frozen bulk prior to formulation and filling, which may take place in a different locality/facility for a variety of reasons. This is not in scope for this standard.

Note that pre-use post installation integrity testing (PUPSIT) may be required for sterilizing filtration which may place challenges to equipment design.